WIX1001
2025/2026 Semester 1
Test 1

PART A

Which of the following statements is *true*?

- C. Every irrational number is a rational number.
- D. Every real number is an imaginary number.
- B. Every rational number is an integer.
- A. Every integer is a rational number.

Two numbers have an HCF of 4 and an LCM of 60. If one of the numbers is 12, find the other.

- 20
- 18
- 15
- 25

Solve the following Linear Equations using Addition / Elimination method:

$2x - y = 1, 3x + y = 14$

- x=4, y=2
- x=2, y=15
- x=3, y=1
- x=3, y=5

Recall the topic of Linear Equations. If y=3x−5 and 2x+y=10, what is the value of x?

- 5
- 2
- 4
- 3

Which of the following pairs of numbers has a Greatest Common Divisor equal to 1?

- 9 and 15
- 16 and 24
- 14 and 25
- 8 and 12

Which of the following propositions is logically equivalent to the expression below:

$$\neg p \rightarrow (q \wedge \neg r)$$

- $p \wedge \neg q \vee \neg r$
- $\neg(p \wedge q) \vee r$
- $\neg p \vee q \wedge \neg r$
- $p \vee (q \wedge \neg r)$

Determine the truth value classification for the compound proposition:

$$(p \wedge q) \wedge \neg(p \rightarrow q)$$

- Logically Equivalent to $p \vee q$
- Tautology
- Contingency
- Contradiction

According to De Morgan's Laws, $\neg(p \wedge q)$ is logically equivalent to:

- $\neg p \wedge \neg q$
- $p \vee q$
- $\neg p \rightarrow q$
- $\neg p \vee \neg q$

Which step shows proof by contrapositive

- a. Assume P, derive Q
- b. Assume Q, derive P
- c. Assume ~P, derive ~Q
- d. Assume ~Q, derive ~P

Over Z, evaluate $\forall x(x \text{ is even} \rightarrow x^2 \text{ is even})$

- a. False
- b. True
- c. Equivalent to: All integers are even
- d. Contingent

Given that B(x) represents customer x ordered a Kopi O.

There were 100 customers in the restaurant. At the end of the day, the manager found out that all the customers ordered a Kopi O.

Therefore, he conclude that $\forall x B(x)$. The process of this inference is called _____ .

- a. Existential instantiation
- b. Universal instantiation
- c. Universal generalization
- d. Existential generalization

Given that B(x) represents customer x ordered a Kopi O.

There were 100 customers in the restaurant and all of them ordered a Kopi O since it is the signature beverage of this restaurant. Judy was a customer of the restaurant.

While not meeting directly, the manager can conclude that B(Judy). The process of this inference is called _____ .

- a. Existential generalization
- b. Existential instantiation
- c. Universal generalization
- d. Universal instantiation

Let G= {a, b}, K = {x, y,z} and L = {0, 1}. Find cardinality of (G x K x L)

- 8
- 24
- 12
- 4
- 6

Which of the following set expressions is equivalent to $(A \cap B) \cup (A \cap B') \cup (A' \cap B)$?

- $A \cup B$
- $A \cap B$
- A'
- $A' \cap B'$

Let $A = \{\emptyset, \{1\}, 2\}$ be a set. Which of the following statements is FALSE?

- $\{\emptyset, \{1\}\} \in A$
- $\emptyset \in A$
- $\{1\} \in A$
- $\{\emptyset\} \subseteq A$

Let $U = \{x | x \in \mathbb{N} \text{ and } x \leq 20\}$ be the universal set. Let $A = \{x \in U : (x-3)^2 < 30\}$. What is the cardinality of the power set of A, i.e., $|P(A)|$?

- 2^{10}
- 2^8
- 2^7
- 2^9

Which set expression is equivalent to $(A \cup B \cup C)' \cup (A \cup B)'$?

- $(A \cap B)' \cap C'$
- $A' \cap B'$
- $A' \cup B' \cup C'$
- $A \cap B \cap C$

In a survey of 120 employees, 80 attend the marketing seminar (M), 70 attend the sales seminar (S), and 5 attend neither. How many employees attend **only** the sales seminar?

- 35
- 40
- 45
- 30

Let $A = \{x \in \mathbb{R} : x^2 < 4\}$ and $B = \{x \in \mathbb{R} : |x - 1| \geq 2\}$. Find the set $A \cap B$

- $(-2, -1)$
- $(-2, -1]$
- $(-\infty, -2)$
- $[-1, 2)$

Which of the following expressions is equivalent to $A \cup (A \cap B)$?

- A
- $A \cap B$
- $\emptyset$
- $A \setminus B$

PART B

If x,y, z ∈ ℤ, given that M(x,y,z) : x/y = z

Which of the following is **true**:

- a. ∃x∀y∃z M(x,y,z)
- b. None of these
- c. ∃x∃y∀z M(x,y,z)
- d. ∀x∃y∃z M(x,y,z)
- e. ∃x∀y∀z M(x,y,z)

If x,y ∈ ℝ, given that M(x,y) : x > y

Which of the following is **true**:

- a. None of these
- b. M(4,6)
- c. M(-π, π)
- d. M(6, 4)
- e. M(5,5)

If $x,y \in \mathbb{R}$, given that $M(x,y) : x<y$

Which of the following is **true**:

- a. $M(-\pi,\pi)$
- b. $M(0,0)$
- c. None of these
- d. $M(5,1)$
- e. $M(9,9)$

Which of the following statements is equivalent to proving the statement: "If a number n^2 is odd, then the number n is odd"?

- Proving: "If a number n^2 is even, then the number n is odd."
- Proving: "If a number n is odd, then the number n^2 is odd."
- Proving: "If a number n is even, then the number n^2 is even."
- None of these
- Proving: "If a number n is even, then the number n^2 is odd."

Let P be the statement: "For every integer $n>1$, there exists an integer k such that $n < k < n^2$ and k is a perfect square."

Which of the following statements regarding a proof of P by contradiction is most accurate?

- Assume there exists an integer $n>1$ such that for all integer k, if k is a perfect square, then $k \leq n$ or $k^2 \geq n^2$.
- Assume there exists an integer $n>1$ such that there is no integer k satisfying $n < k < n^2$ and k is a perfect square.
- Assume there exists an integer $n>1$ such that for all integer k, we have $k^2 \leq n$ or $k^2 \geq n^2$.
- none of these

Let

$$P : \text{''I will pass the course''}$$

$$I : \text{''I will get an internship''}$$

$$W : \text{''I work hard''}$$

Which symbolic expression correctly represents the English sentence:

''It is not the case that I will pass the course or get an internship, but if I work hard then I will pass the course.''

- $\neg P \vee I \wedge (W \rightarrow P)$
- $\neg(P \vee I) \wedge (W \rightarrow P)$
- None of these
- $(\neg P \vee \neg I) \vee (W \rightarrow P)$
- $\neg P \wedge \neg I \wedge W \rightarrow P$

Let **A** be "The alarm rings", **B** be "I wake up", and **C** be "I am late". Which symbolic expression correctly translates the statement: "I am not late only if I wake up, and I wake up only if the alarm rings."?

- $(\neg C \leftrightarrow B) \wedge (B \leftrightarrow A)$
- $(\neg C \wedge B) \rightarrow (B \vee A)$
- $(\neg C \rightarrow B) \vee (B \rightarrow A)$
- $(\neg C \rightarrow B) \wedge (B \rightarrow A)$
- None of these

Simplify the compound proposition using the Laws of Logic:

$$(\neg p \wedge q) \vee (\neg p \wedge \neg q)$$

- $\neg p$
- $p \vee q$
- $p \wedge q$
- None of these
- $\mathbf{F}$ (Contradiction)

Consider the argument:

"If the product is defective (p), I will get a refund (q). I did not get a refund ($\neg q$). Therefore, the product wa

This argument is an example of which valid rule of inference or common fallacy?

- Modus Tollens
- None of these
- Hypothetical Syllogism
- Affirming the Consequent

Analyze the validity of the argument below:

Premises:

$$p \vee q$$

$$p \to r$$

$$q \to s.$$

Conclusion:

$$r \vee s$$

.

- The argument is **Valid** only if r and s are logically equivalent.
- The argument is **Invalid** because the conclusion is a disjunction, which is not guaranteed by the conditional premises.
- The argument is **Invalid**; it is an example of the Fallacy of Denying the Antecedent.
- The argument is **Valid** because if the premises are true, the conclusion must also be true.

The original conditional statement is:

$$\text{A necessary condition for the car to start } (P) \text{ is that the battery is charged } (Q).$$

Which of the following represents the **Inverse** of this statement?

- If the battery is charged, the car will start.
- If the car does not start, then the battery is not charged.
- None of these
- If the car starts, then the battery is charged.
- If the battery is not charged, the car will not start.

The following propositions are being compared for logical equivalence:

1. $\neg p \vee q$
2. $p \rightarrow q$
3. $\neg q \rightarrow \neg p$
4. $q \rightarrow p$

Which of the following sets contains ALL propositions that are logically equivalent to each other?

- The set $\{1, 3, 4\}$
- The set $\{1, 2, 4\}$
- None of these
- The set $\{2, 3, 4\}$
- The set $\{1, 2, 3\}$

Which shows a counterexample method?

- a. Prove ∀xP(x) by cases
- b. Disprove ∃xP(x) by contradiction
- c. Prove ∃xP(x) by showing P(x) holds ∀x
- d. Disprove ∀xP(x) by exhibiting a specific a with ~P(a)
- e. None of these

Let $A = \{x \in \mathbb{Z} : x^3 = 8\}$. Let $B = \{x \in \mathbb{N} : x < 3\}$. What is the cardinality of the Power Set of $A \times B$?

- 2
- 2^3
- None of these
- 2^6
- 4

In a class of 80 students, 35 took Art (A), 42 took Biology (B), and 30 took Chemistry (C). 10 students took all three subjects. 15 took both A and B. 13 took both B and C. 12 took both A and C. How many students took exactly one of the three subjects?

- 60
- 47
- 52
- 54
- None of these

If x refers to everyone, ∀x(S(x) → K(x)) can be used to represent which sentence?

- a. Everyone is smart and likes karaoke.
- b. If Donald is smart, he will lives in Kuala Lumpur.
- c. All people are smart if they like karaoke.
- d. Those who are smart live in Kuala Lumpur.
- e. None of these.

Given x refers to everyone, P(x): x is happy, Q(x): x is rich

∀x(P(x)→¬Q(x)) represents _____ .

- a. Everyone is happy and rich.
- b. None of these.
- c. All happy people are not rich.
- d. All rich people are happy.
- e. A person named x is happy and rich.

Translate the following sentence to symbolic form:

"No surface that is sanitized is unclean."

- a. $\forall x(\neg Sanitized(x) \rightarrow Clean(x))$
- b. $\neg \exists x (Sanitized(x) \wedge Unclean(x))$
- c. $\neg \forall x (Sanitized(x) \wedge Unclean(x))$
- d. None of these
- e. $\exists x (Sanitized(x) \wedge Clean(x))$

Translate the following sentence to symbolic form:

"Some successful people are not lucky"

x refers to everyone.

- a. $\exists x (Successful(x) \wedge Lucky(x))$
- b. $\forall x (Successful(x) \wedge \neg Lucky(x))$
- c. $\exists x (Successful(x) \wedge \neg Lucky(x))$
- d. None of these
- e. $\forall x (Successful(x) \wedge Lucky(x))$

Translate the following sentence to symbolic form:

"There is no free lunch."

Given that:
F(x) represents "x is free"
L(x) represents "x is lunch"

- a. $\exists x (F(x) \wedge L(x))$
- b. $\sim \exists x (F(x) \wedge L(x))$
- c. none of these
- d. $\forall x (F(x) \rightarrow L(x))$
- e. $\sim \forall x (F(x) \rightarrow L(x))$